Tanmoy Sen Gupta

Lead Software Engineer at Optum - a UnitedHealth Group company
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Summary

Lead Software Engineer at Optum (UnitedHealth Group), focused on enterprise data infrastructure at scale. Works across the full stack to architect, design, and develop UHG's Enterprise Data Catalog and Access Provisioning platform, delivering reliable, high-impact data solutions for one of the largest healthcare organizations in the world.

Experience

Optum July 2021 – Present
Lead Software Engineer *Hyderabad*

- Architecting and developing UHG's Enterprise Data Catalog and Access Provisioning platform, incorporating AI-assisted workflows to improve data discoverability, access management, and scalability.
- Automated and standardized internal processes and governance workflows, including approval automation and policy management services, significantly reducing manual support effort
- Built scripts, visualization tools, and proxy endpoints to model governance flows and simplify management of data products and entities, improving transparency and operational efficiency
- Collaborated with product owners on requirements and UI/UX design, while contributing to distributed system design and cross-functional delivery of enterprise data platform capabilities
- Previous roles: Senior Software Engineer, Software Engineer

AstraZeneca June 2021 – July 2021
Trainee *Chennai*

Education

SRM Institute of Science and Technology India
Bachelor of Technology in Computer Science and Engineering *2017 – 2021*

- Focused on theoretical computer science and cryptography, with a final year project building hyENC3 — an advanced multi-layer hybrid cryptography system for securing sensitive data. CGPA: 7.89.
- Organized and designed branding collateral for large-scale college Fests and Model United Nations (MUN) events, contributing to end-to-end event planning and coordination.

Technical Skills

Languages: Golang, Python, Java, TypeScript
Backend & APIs: REST, Microservices, Distributed Systems
Cloud & DevOps: Kubernetes (AKS, GKE), Docker, CI/CD, GitHub Actions
Databases: MongoDB, NoSQL, Data Modeling

Open-Source Projects

wunderDB | *Go, Docker, Kubernetes* | github.com/TanmoySG/wunderDB 2020 – Present

- Designed and built a JSON-based in-memory data store inspired by MongoDB with a RESTful API.
- Developed `wdbctl`, a Homebrew-installable CLI tool for managing wunderDB instances.
- Maintained Go client library (`wdb-go`) and interactive dashboard (`wunderDash`) for the platform.

go-steps | *Go* | github.com/TanmoySG/go-steps 2023 – Present

- Created a Go library enabling developers to define, execute, and manage functions as sequential step-chains.
- Implemented conditional branching, configurable retry logic with error pattern matching, and sleep intervals.
- Built built-in rollback functionality with `RollbackFn` for automatic cleanup on step failure.
- Integrated zerolog-based structured logging with per-step progress tracking via `GoStepsCtx`.

Find more projects on my Github.

Certifications

GitHub Foundations, issued by GitHub
Microsoft Certified: Azure Fundamentals, issued by Microsoft
AI Dojo: Introduction to building AI Systems, issued by Optum

Publications

Advanced Data Security using multiple cryptographic algorithms over multiple layers and secured key sharing
Jan 2021 – Apr 2021
Security of data is one of the most important aspects of digital presence. This work proposes a multi-layered encryption protocol using AES, RSA, and DES in a cascading manner, with secured key sharing after authentication to address weaknesses in standalone encryption systems.
Tech: Python, Cryptography

Show project | **Show Paper**

Interests & Community

Open-Source: Actively maintains multiple open-source Go and Python libraries used by the developer community.
Technical Writing: Publishes engineering articles and guides at blog.tanmoysg.com.